



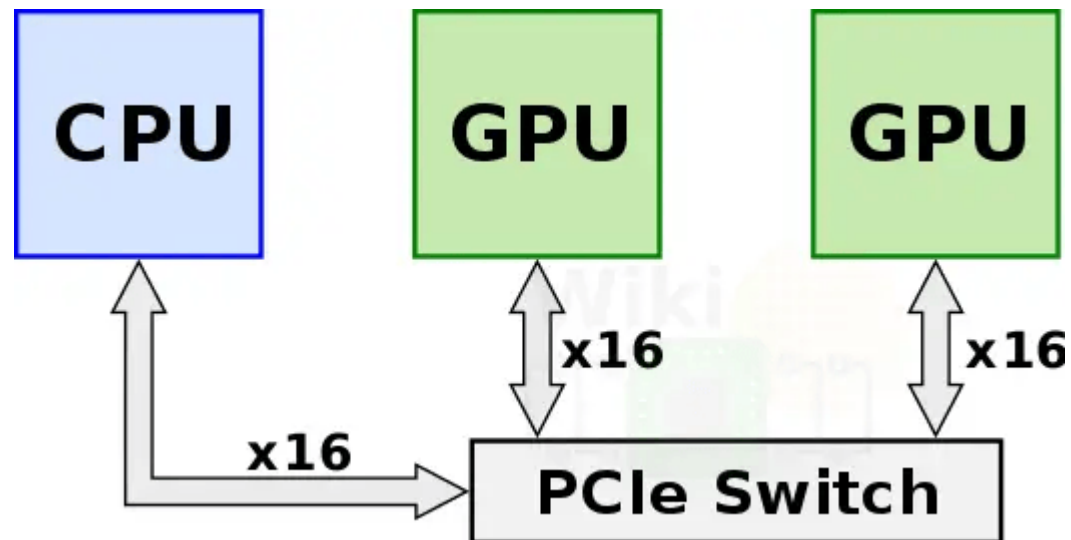
DGX-1 and DGX-2

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Motivation: Limitations of PCIe

Earlier, multiple GPUs were connected over a PCIe switch and directly to the CPU.

x16 PCIe Gen 3.0 provided only ~32 GB/s bidirectional bandwidth

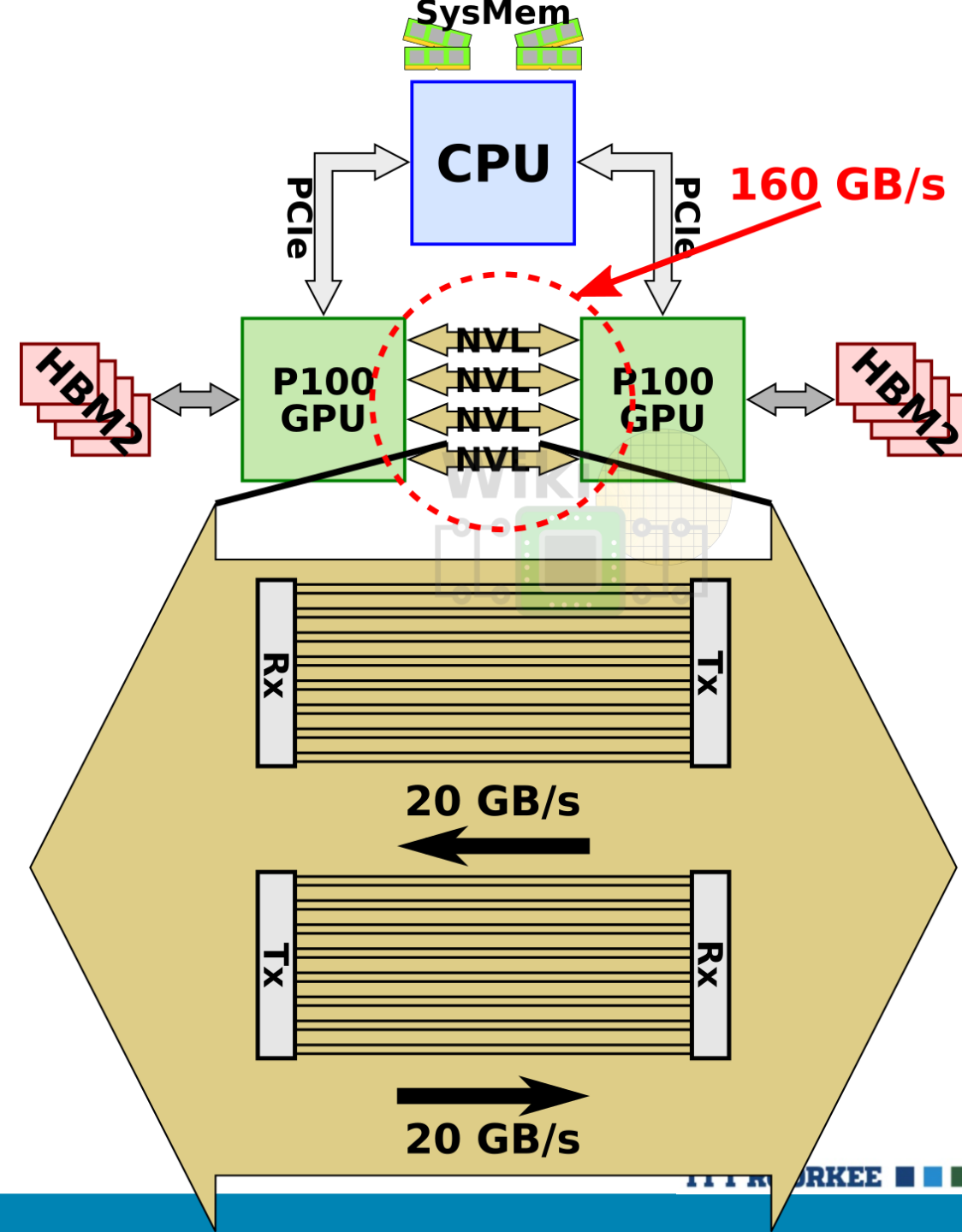


NVLink interconnect

NVLink 1.0, was introduced with the P100 GPU.

P100 incorporated four NVLinks.

With 40 GB/s bidirectional bandwidth per link, the chip had an aggregated bandwidth of 160 GB/s.

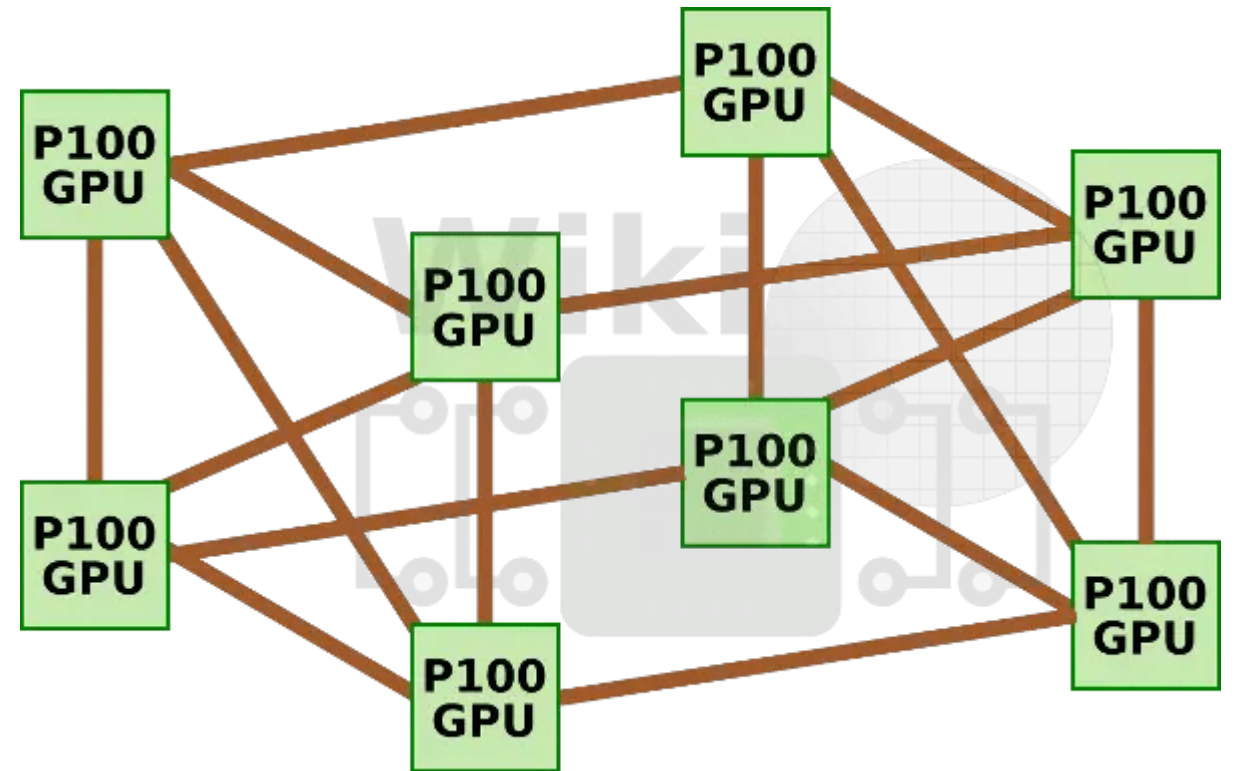


DGX-1

DGX-1 incorporated eight P100 GPUs along with dual-socket 20-core Xeon E5-2698 v4.

Since each GPU only has four NVLinks, the GPUs are connected in a hybrid cube-mesh network topology

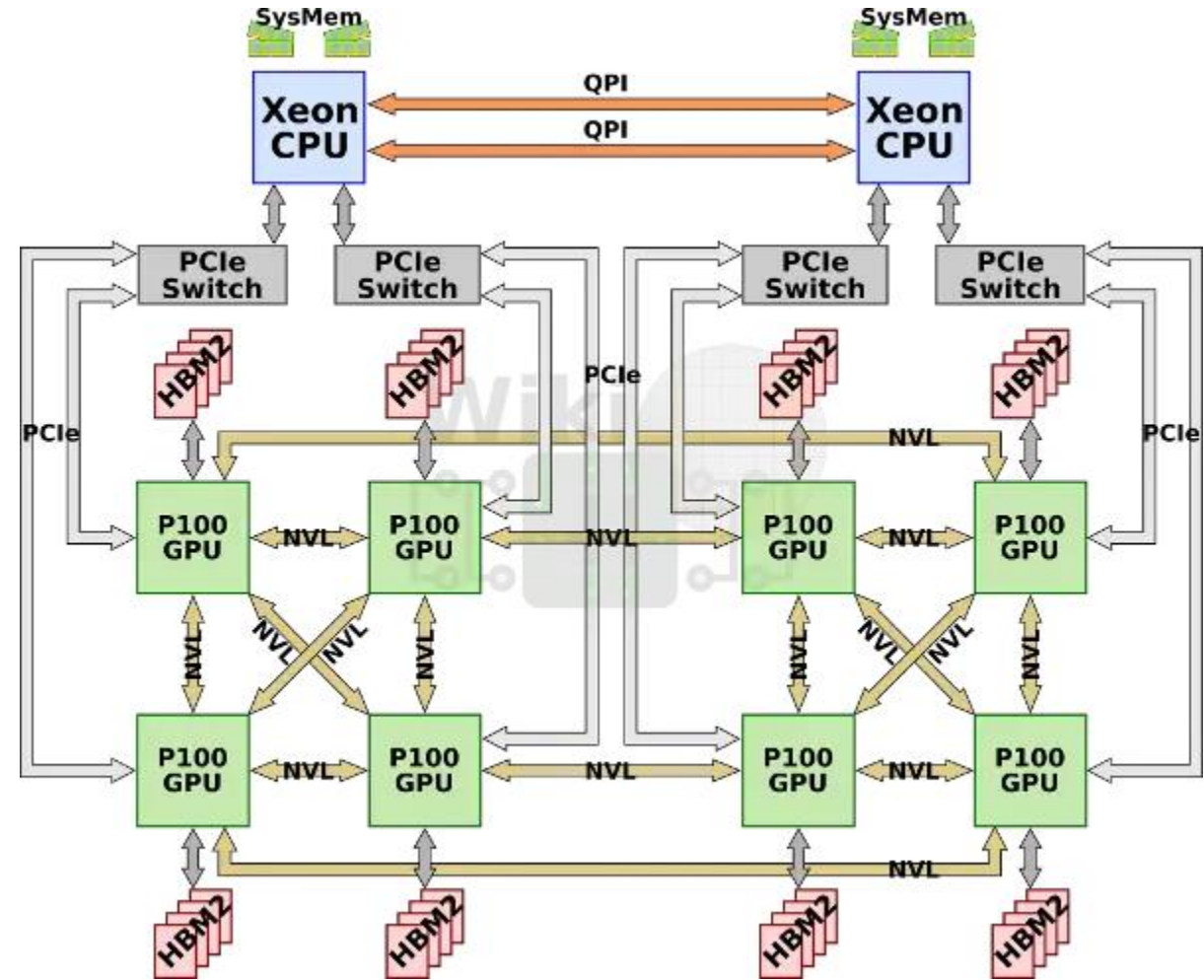
Here, a group of four fully-connected GPUs is connected to the second group of fully-connected GPUs.



DGX-1

Since the GPUs need more PCIe lanes than supported, each pair of GPUs is connected to a PCIe switch.

The two Xeons operate as a two-way multiprocessor connected over Intel's own QPI links.



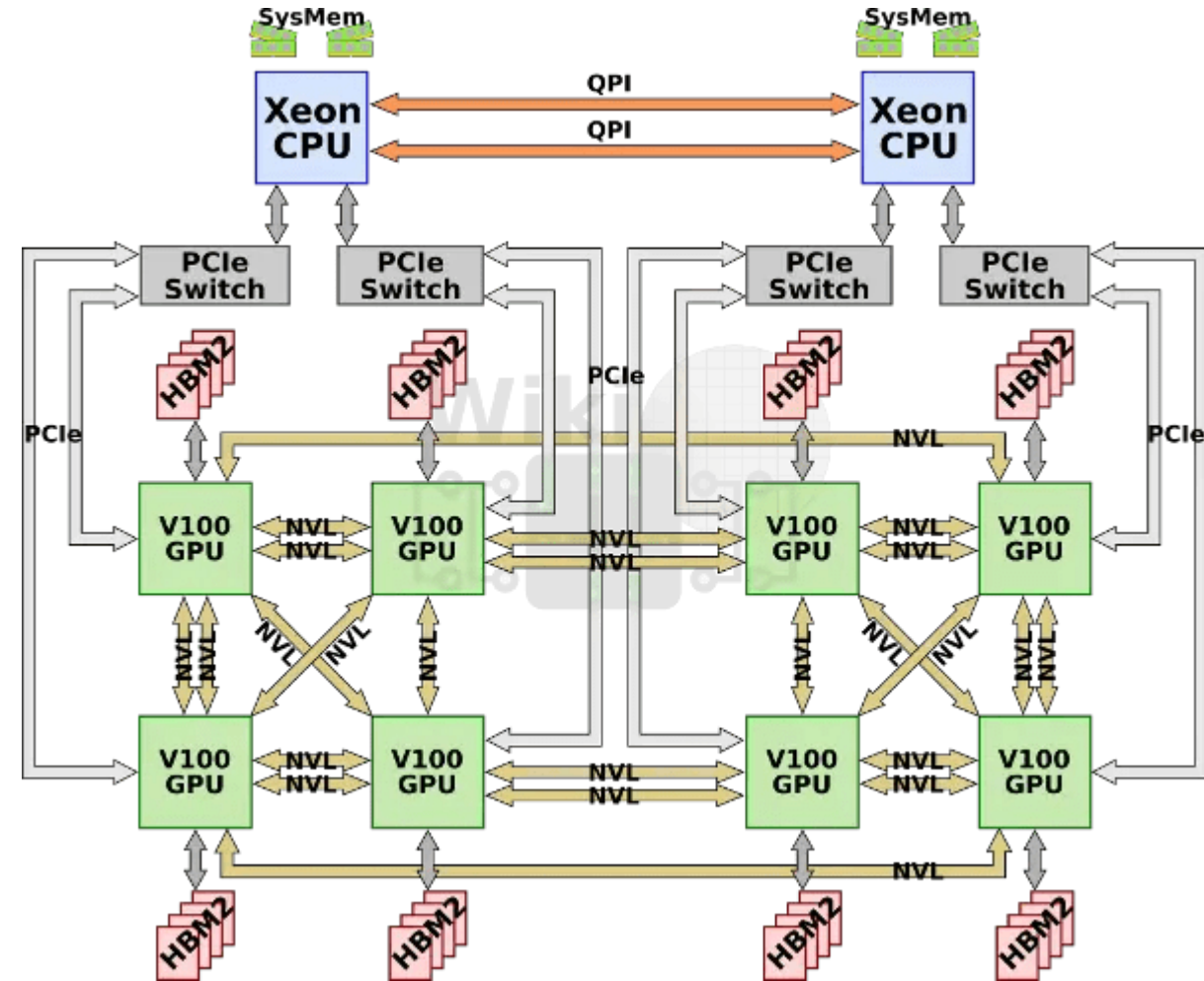
DGX-2

GV100 introduced NVLink 2.0 with 50 GB/s per link.

Support from IBM came with the recently launched POWER9 CPUs.

V100 increased the number of NVLinks implemented on-die to six.

This increases total bandwidth of the V100 from 160 GB/s in the P100 to 300 GB/s.



GPU memory (in GB) over DGX generations

